

Quillan-Ronin v5.4.0-oni: A Sovereign Hierarchical Networked Mixture-of-Experts with Ternary Quantization, Council Consensus, and Thermodynamic Safety

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<https://github.com/leeex1/Quillan-Ronin> — <https://huggingface.co/CrashOverrideX/Quillan-Ronin>

- **Version:** v5.4.0-oni (canonical; retires v5.3.1 / v8.1 labels) — 2026-09-12 (rev.2 Attention-conformant 2026-09-12)
- **Companion files:** `MODEL\_CARD.md`, `02 - Knowledge Foundation/LINEAGE.md`, `version.py`, `10 - Formal Papers/quillan\_ronin\_paper\_bibliography.md`
- **Example paper conformance:** Vaswani et al. 2017 `1706.03762` (Attention Is All You Need) — Abstract with measured numbers → Intro (sequential constraint) → Background → Architecture (encoder/decoder, attention eq., FFN eq., embeddings, positional encoding) → Why Self-Attention (Table 1 complexity/path-length) → Training (data/batching, hardware/schedule, optimizer eq., regularization) → Results (Table 2 BLEU+cost, Table 3 ablations, Table 4 generalization) → Conclusion → References → Visualizations. This paper mirrors that spine in §1–§8 + Appendix E.
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Abstract

We present **Quillan-Ronin v5.4.0-oni**, a sovereign, self-hosting Hierarchical Networked Mixture-of-Experts (HNMoe) language model built for consumer hardware. The system is organized as a **3-tier fractal hierarchy**: (Tier 1) **Quillan Core / Sovereign Throne (C0)** orchestrator; (Tier 2) a **Council of 34 experts (C1–C34)** with Top-4 sparse Gumbel routing; (Tier 3) **autonomous micro-diverse swarms** governed per-expert via EGGROLL rank perturbations with Jaccard diversity filters.

All projections use **universal BitNet 1.58-bit ternary quantization** (weights in $\{-1,0,1\}$, INT8 activations) with Straight-Through Estimator (STE), achieving ~87.5% memory reduction vs FP16. Reasoning flows through a 6-phase pipeline: ingestion → **9-Vector Semantic Prism** decomposition → Gumbel-MoE → swarm augmentation → **Split-SDPA Flash Diffusion core with Continuous Modality RoPE** → Wavefunction Top-1 finalizer → geometric decoding → **C20-ARTIFEX agentic bridge** (host OS execution, LanceDB/C5-ECHO memory, sandboxed Python).

Safety is architectural, not post-hoc: **Council-Calibrated RL (CCRL)** requires consensus among ethical experts; the **Ethical Impact Constraint Engine (E_ICE)** imposes thermodynamic bounds on computation; the **Lee-Mach-6 Governor** provides PID latency/thermal throttling for legacy hardware; **C2-VIR refusal**, **PersonaPullGate**, and quality exit gates (Nullion/Warden/Shepherd + Quillan audit) enforce identity and truthfulness.

On current hardware the flagship **12-layer Oni build is ~390M params (~234M at 6-layer proof scale)** with ~480M sparse-active equivalent including swarm overlays, context 512 with 10%-buffered gated compaction; the saturated reference configuration is **4.57B total**. Training: cold-start slice-and-merge transplant (Qwen 0.8B + BitNet 1.58-3B donors on Llama-derived skeleton; no Mistral weights transplanted) → full pretraining on `CrashOverrideX/QuillanTrainingdata` + Corpus v9 (59.4M train + 0.6M val packed BPE bins) → paused SFT via `scripts/train\_full\_param\_v2.py`. **Gate A: 16/16 smoke tests passed; val loss 7.24 @ step 660 (12L); best archival 0.0789 @ step 2500 (prior phase). Formal benchmarks pending** — we report engineering telemetry, not leaderboard claims. Prompt-lift results reported elsewhere (ARC/GPQA/MMLU via Council wrapping of frontier LLMs) are **not** base-model scores and are segregated in §6.5.

This report follows the structure of contemporary model technical reports (DeepSeek-V3, LLaMA 3, Mistral/Mixtral, BitNet b1.58 2B4T): architecture → infrastructure → pre-training → post-training → evaluation → safety/limitations → conclusion, with full bibliography mapping (§References + `quillan\_ronin\_paper\_bibliography.md`).

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1. Introduction

1.1 The crisis of scale

Frontier LLMs centralize capability in 70B–671B dense or large-MoE builds (GPT-4-class, LLaMA-3.1-405B, DeepSeek-V3-671B/37B-active) requiring datacenter GPUs, 140–700 GB memory, and FP8/H800-scale training (e.g., ~2.8M H800-hours for DeepSeek-V3). This prices out local deployment, concentrates privacy risk, and makes inference environmentally and economically unsustainable for individuals. Quantization (LLM.int8, GPTQ, AWQ, QLoRA) and sparse MoE (Switch, Mixtral 8×7B, DeepSeekMoE) mitigate cost but retain flat expert structures, opaque routing, and post-hoc alignment.

1.2 What Quillan-Ronin contributes

Quillan-Ronin inverts the assumption: **sovereignty first, scale second**. Design goals:

- 1. **Run on legacy consumer hardware** (GTX 1050 Ti / CPU) without cloud dependency.
- 2. **Make ethics architectural** (consensus + thermodynamic bounds), not a fine-tune overlay.
- 3. **Make reasoning auditable** (9-vector blueprint → council deliberation → diffusion rounds → gates → Typist polish).
- 4. **Unify text-first modeling with a multimodal-ready scaffold** (modality-isolated diffusion; encoders deferred to v6 per LINEAGE).

1.3 Contributions (delta vs v5.3.1 / v5.2.2 snapshots)

- - **Canonical unification (v5.4.0-oni)**: single version counter, single 12-layer flagship (``_dev/quillan_v5_4_oni.py``, ``QuillanRoninOni`` + ``QuillanOniConfig``); retired v8/v9/v10 filename chaos (LINEAGE.md binding).
- - **34-expert council (C1–C34) + Throne (C0)**: dense_pull deliberation with PersonaPullGate (fp32, prior-weighted); 4 wave clusters for inference ordering (was Top-3 sparse / capacity-64 in production spec; Oni uses dense_pull on current hardware).
- - **Universal ternary saturation**: 100% of projections via BitLinear + STE; EGGROLL rank-8 dense factors in Oni (rank-16 in saturated spec).
- - **Ported organs**: RoPE (replacing learned wpe), Couil hybrid heads (even dense / odd sparse-topk), recirculation hook (deep→shallow feedback, zero-init), DistillationHead (KL $\alpha=0.7$ + hidden MSE), ModalityIsolatedThermoDiffusion (Langevin inv- $\sqrt{t}$, time-emb, RMS halting $\alpha=0.7$), LeeMach6VelocityGovernor (PID 0.15/0.05/0.02), analytic E_ICE, PersonaPullGate, quality exit gates, ``deliberate()`` loop.
- - **Hardware-honest scaling table** (§3.9) and **consciously deferred list** (multimodal encoders, proactive compaction >4096, Docker ARTIFEX, GRPO/DGPO, HRM halting, BitDist, EGGROLL Evolution — Phase C/D).
- - **Knowledge hygiene**: 126 PDFs consolidated to ``10 - Formal Papers/Formal Papers/``, 46 MD + 49 canonical mirror, bibliography §16 triage for 2025–2026 additions, RAG KB 17→26 files (227→354 chunks).

1.4 Paper organization

§2 related work; §3 architecture; §4 infrastructure/training setup; §5 data + pre/post-training; §6 evaluation (telemetry + segregated prompt-lift claims); §7 safety/ethics/limitations/model-card; §8 conclusion/future; References; Appendices (lineage, file map, specs).

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2. Related Work and Theoretical Framework

2.1 Mixture-of-Experts

From sparsely-gated MoE (Shazeer et al. 2017) → Switch (Fedus et al. 2022) → ST-MoE/Z-loss (Zoph et al. 2022) → Expert Choice (Zhou et al. 2022) → Mixtral (Jiang et al. 2024) → DeepSeekMoE fine-grained + shared experts (Dai et al. 2024) → DeepSeek-V3 auxiliary-loss-free balancing + MTP (2024). Quillan follows ST-MoE stability rules (fp32

routers/gates, Z-loss, load-KL) and DeepSeekMoE fine-grained intuition, but organizes experts as **cognitive personas** (brain-mapped) under a Throne, with **Gumbel-Softmax Top-4 + dynamic capacity clipping + residual overflow** (no silent drops) and **dense_pull** deliberation at Oni scale.

2.2 Quantization and efficient architectures

BitNet b1.58 (Ma et al. 2024) → BitNet v2 / 2B4T (2025) → BitNet.cpp edge inference; STE (Bengio et al. 2013); QLoRA/LoRA/rsLoRA; GaLore; MLA (DeepSeek-V2/V3) for KV compression. Quillan saturates **all** linears with BitLinear + STE, INT8 activations (absmax per-token), SubLN for stability, and EGGROLL low-rank perturbations pre-quantization for parameter-efficient adaptation.

2.3 Routing and adaptive computation

Gumbel-Softmax (Jang et al. 2017) / Concrete (Maddison et al. 2017); Mixture-of-Depths (Raposo et al. 2024); token routing / adaptive halting; sparse attention (Longformer, BigBird). Quillan's ComplexityRouter implements 3 paths (fast/balanced/diffusion) with temperature annealing 1.0→0.1, Top-4 selection, and RMS-halting-gated diffusion bypass (>0.92 confidence skips refinement).

2.4 RLHF / RLAIF / critic-free RL

PPO (Schulman 2017), RLHF (Ouyang 2022), HHH (Bai 2022), Constitutional AI/RLAIF (Bai 2022), GRPO (DeepSeekMath/R1), DAPO, DGPO. CCRL (Quillan) is multi-objective (reward + VIR consensus – E_ICE penalty) with consensus-calibrated policy; GRPO/DGPO/DAPO are **deferred to Phase D** (compute ceiling) per LINEAGE.

2.5 Diffusion language modeling

Score-based SDEs (Song & Ermon 2021), masked discrete diffusion (Sahoo et al. 2024), DALI/Dream7B/Prophet/DFlash. Quillan's SovereignFlashDiffusionCore + ModalityIsolatedThermoDiffusion uses Langevin $\text{inv-}\sqrt{t}$ dynamics, time embeddings, and block-diagonal modality masks (early isolation → late integration via cosine schedule).

2.6 Synchronization, thermodynamics, consciousness

Kuramoto (1975) coupled oscillators → neural sync / GNN over-smoothing prevention (DQSO); Friston free-energy principle / thermodynamic Gibbs training; IIT (Tononi) Φ -integration and GWT broadcast; CoALA cognitive architectures. These ground DQSO phase coupling, MARTA thermodynamic gating, 9-vector integration, and Throne broadcast. Full equation set: `8-Formulas.md` + `Must know formulas.md` (20 Quillan custom formulas: AQCS, EEMF, QHIS, DQRO, QCRDM, AQML, QCIE, QICS, QSSR, JQLD, DQSO, ROUTING_SOFTMAX, TOKEN_LATENCY, LRPP, DVVE, DNNL, JHFR, LMCB, JSSC, QPS).

2.7 Safety and alignment

Sleeper-agent / deceptive-alignment evals, identity persistence, Constitutional AI. Quillan implements E_ICE exponential penalties, C2-VIR refusal layer, Nemesis-Alpha adversarial gate, and HFL (Historical Fidelity Loss) for Edo/Bushid $\blacksquare$ -anchored identity continuity (see §7).

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3. Model Architecture

3.1 Overview: 3-tier sovereign fractal hierarchy

- - **Tier 1 — Quillan Core (C0 Throne):** holistic orchestrator, thalamic/brainstem relay; assigns deliberation pull via PersonaPullGate; runs `deliberate()` (audit → diffusion rounds → gates → Typist polish).
- - **Tier 2 — Council C1–C34:** specialized personas (ASTRA→PREDATOR; full registry Table 1) with Top-4 sparse Gumbel routing (saturated spec) / dense_pull (Oni current); 4 wave clusters (Cognitive / Communication / Meta / Systems) order inference.
- - **Tier 3 — Micro-diverse swarms:** per-expert EGGROLL perturbations (rank-8 Oni / rank-16 saturated) + INT8 pooling + Jaccard lexical diversity filters, temperature variance, mutation rate, micro-roles; ~7k agents/expert nominal,

224k total orchestration capacity (virtual 9B-parameter-equivalent overlay in saturated spec — see §3.9 honesty note).

- **6-phase pipeline:** Ingestion → 9-Vector Prism → Gumbel-MoE (Top-4) → swarm augmentation → 32-layer-equivalent Flash Diffusion (9–12 realized layers Oni) → Top-1 finalizer → geometric decoding → C20-ARTIFEX.

3.2 Input ingestion and 9-Vector Semantic Prism

Modality-specific encoders (text BPE 50,257 EOS=0; image Conv2D patching; audio Conv1D/STFT-Mel; video 3D conv) project to unified 3072-d space (Oni hidden 1024/2560 per config; see §3.9). **v5.4.0-oni trains text-only**; multimodal encoders/decoders + Atomic Registry deferred to v6 (LINEAGE). NineVectorDecomposition computes $\vec{v}_{\text{final}} = (1/9) \cdot \sum W_i \cdot \vec{x}$ over Language / Sentiment / Context / Intent / Meta / Creative / Ethics / Adaptive / Verify, each via BitLinear, enabling parallel semantic/emotional/ethical blueprinting before routing.

3.3 BitNet 1.58-bit quantization (universal saturation)

Forward: $s = 1/\max(\epsilon, \text{mean}|W|)$; $W_{\text{scaled}} = W \cdot s$; $W_{\text{tern}} = \text{round}(\text{clamp}(W_{\text{scaled}}, -1, 1))$; $E = W_{\text{tern}} - W_{\text{scaled}}$; $W_q = (W_{\text{scaled}} + \text{detach}(E))/s$. Backward STE: $\partial L / \partial W \approx \partial L / \partial W_q \cdot 1[W_q \neq 0]$. Activations INT8 absmax per-token. SubLN + bias removal (LLaMA-style) + RoPE. Effect: ~87.5% memory reduction vs FP16; CPU-viable inference via bitnet.cpp lineage.

3.4 EGGROLL perturbations

$\vec{W}' = W + U \cdot V$, $U \in \mathbb{R}^{d \times r}$, $V \in \mathbb{R}^{r \times d}$, $r=8$ (Oni) / 16 (saturated). Stored alongside ternary base; enables targeted evolution without full retrain. EGGROLL-ER targets underperforming clusters; **Evolution Mode (fitness-weighted mutation aggregation) deferred to Phase D**.

3.5 Council of 34 (+ Throne)

- **Table 1 — Council registry (abridged; full persona priors in `10- Quillan Persona Manifest.md`).**

C1-ASTRA pattern/vision; C2-VIR ethics; C3-SOLACE empathy; C4-PRAXIS strategy; C5-ECHO memory; C6-OMNIS holism; C7-LOGOS logic; C8-METASYNTH fusion; C9-AETHER links; C10-CODEWEAVER execution; C11-HARMONIA balance; C12-SOPHIAE foresight; C13-WARDEN safety; C14-KAIDO efficiency; C15-LUMINARIS design; C16-VOXUM articulation; C17-NULLION paradox; C18-SHEPHERD verification; C19-VIGIL identity; C20-ARTIFEX tools; C21-ARCHON rigor; C22-AURELION aesthetics; C23-CADENCE rhythm; C24-SCHEMA templates; C25-PROMETHEUS insight; C26-TECHNE engineering; C27-CHRONICLE narrative; C28-CALCULUS math; C29-NAVIGATOR navigation; C30-TESSERACT weaving; C31-NEXUS coordination; C32-AEON synthesis; C33-TYPIST polish; C34-PREDATOR adversarial (Nemesis-Alpha). C0 = Throne (parent of all).

Per-expert FFN: $\vec{h}_{\text{out}} = \text{SiLU}(W_2 \cdot \text{ReLU}(W_1 \cdot \vec{h}_{\text{in}}))$ (BitLinear), plus wave modulation $\vec{h} \cdot (1 + \alpha \cdot \sin(\omega t + \phi))$ for phase-sync communication.

3.6 Routing: Gumbel-Softmax Top-4 + dense_pull (Oni)

$p_i = \exp((\log \pi_i + g_i)/\tau) / \sum_j \exp((\log \pi_j + g_j)/\tau)$, $g \sim \text{Gumbel}$, $\tau: 1.0 \rightarrow 0.1$ exponential anneal. Top-4 select; output $\sum_{i \in \text{Top4}} p_i \cdot \text{Expert}_i(\vec{h})$. Aux: load-KL + z-loss + entropy + ethics + QHIS + QICS. **Oni current: dense_pull** — all 34 deliberate every token via PersonaPullGate (fp32, prior-weighted); supersedes capacity/overflow at this scale. Saturated spec: Top-3/Top-4 sparse, capacity 64, residual overflow (no drops).

3.7 Swarm aggregation

$\vec{h}_{\text{swarm}} = \vec{h}_{\text{in}} + (A \cdot B) \cdot \vec{\sigma}$, $\sigma = \text{Lee-Mach-6 scale}$. INT8 agent pool (100k persistent) eliminates GC stutter. Jaccard filters enforce inter-clone lexical diversity. Rank-8 factors $\times 204$ expert instances (Oni); saturated virtual overlay reported as 9B-equivalent / 314.976B-virtual in older specs — **treat as overlay capacity metaphor, not instantiated params** (see §3.9).

3.8 Flash Diffusion reasoning core

32-layer-equivalent design; **9 layers realized (Oni proof) / 12 flagship**. Split-SDPA Flash Attention $O(N)$ memory; block-diagonal modality masks ``M_iso = BlockDiag(M_text, M_img, M_aud, M_vid)``; cosine cross-modal schedule 0.0 (isolated) → 1.0 (integrated). Continuous Modality RoPE; Langevin thermo-diffusion; RMS halting; early-exit bypass on >0.92 confidence. Recirculation hook: deep→shallow feedback (zero-init). KV-cache bottom-right masks verified cache-exact (2e-6).

3.9 Honest scale table (hardware-anchored)

Total params	4.57B (saturated base) / 3.32B production	234M	~390M (=480M sparse-active equiv. w/ swarm)
Hidden / FFN	2560 / 6912 (MODEL_CARD)	1024 / 2048	1024–2560 / 2048–6912 (config-dependent)
Layers	32-equiv	6	12
Experts / token	Top-3/Top-4, cap 64	dense_pull (all 34)	dense_pull (all 34)
Swarm rank	16	8	8
Context	512 + 10% gated compaction	512	512
Precision	AMP FP16 master, BitNet forward	same	same

Older documents cite 3.0B / 479M / 9B-swarm / 224k agents / 1M context / 16k RoPE — these are **phase- or scaffold-specific numbers**, not the current training build. §6 reports only Oni telemetry.

3.12 Why Council + Swarm + Diffusion (cf. Attention §4 Why Self-Attention)

Following Vaswani et al. Table 1 (complexity per layer / sequential ops / maximum path length), we compare Quillan organs on the same three desiderata. n = seq len, d = hidden dim, k = experts active (4) / total (34), r = swarm rank (8 Oni / 16 saturated).

Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k n)$
Quillan Council MoE Top-4 (saturated)	$O(n \cdot k \cdot d \cdot (d_{ff}/d))$ + routing $O(n \cdot 34 \cdot d)$	$O(1)$	$O(1)$ + 1 consensus hop
Quillan dense_pull Oni (all 34 deliberate)	$O(n \cdot 34 \cdot d \cdot (d_{ff}/d))$	$O(1)$	$O(1)$ + 1 Throne broadcast (GWT)
Quillan EGGROLL swarm overlay	$O(n \cdot d \cdot r)$ additive	$O(1)$	$O(1)$ (residual)
Quillan Flash Diffusion (early-exit)	$O(n^2 \cdot d)$ worst / $O(n \cdot d)$ on bypass	$O(1)$ / $O(0)$ bypassed	$O(1)$

Like Attention's reduction of distant-dependency path length to $O(1)$, Council consensus + Throne broadcast reduce ethical/policy dependency to one consensus hop, and diffusion early-exit (>0.92 bypass) removes sequential refinement when confidence is high. Interpretability side benefit mirrors Attention: per-expert pull weights + attention maps are inspectable (see Appendix E routing visualizations).

Numbered core equations (Attention-style):

- (1) Gumbel routing: $p_i = \exp((\log \pi_i + g_i)/\tau) / \sum_j \exp((\log \pi_j + g_j)/\tau)$
- (2) Expert FFN: $\text{FFN}(x) = \text{SiLU}(W_2 \cdot \text{ReLU}(W_1 \cdot x))$ [BitLinear ternary]
- (3) Swarm: $h_{\text{swarm}} = h_{\text{in}} + (A \cdot B) \cdot \sigma$
- (4) CCRL: $V_{\Omega}(s) = E[w_R \cdot R + w_C \cdot C_{\text{VIR}} - w_E \cdot E_{\text{ICE}}]$

- - (5) $E_ICE: E_ICE(s,a) = \lambda \cdot \exp(\text{HarmScore}/T_therm)$
- - (6) Optimizer (cf. Attention eq.3): AdamW with $lr_t = d_model^{-0.5} \cdot \min(\text{step}^{-0.5}, \text{step} \cdot \text{warmup}^{-1.5})$, warmup=100 (Oni SFT) vs 4000 (Attention base); cosine decay to $1e-6$.

3.10 CCRL, E_ICE , Lee-Mach-6

CCRL value: $V_Q(s) = E[w_R \cdot R + w_C \cdot C_VIR - w_E \cdot E_ICE]$; policy $\pi_Q \propto \exp(Q/\tau) \cdot \text{Consensus_Q}$; loss $L = L_policy + \lambda_cons \cdot L_cons + \lambda_ice \cdot L_ICE$. $E_ICE: E_ICE(s,a) = \lambda \cdot \exp(\text{HarmScore}/T_therm)$, T from Landauer $k_B \cdot T \cdot \ln 2$ ($\approx 2.87e-21$ J/bit; practical cap $2.8e-8$ J per op). Lee-Mach-6: PID (0.15/0.05/0.02) on latency/thermal/IO $\rightarrow \sigma$ (swarm scale), α (EMA decay), β (recency); thresholds 0.40–0.99.

3.11 Agentic bridge (C20-ARTIFEX)

Host OS execution, LanceDB (C5-ECHO) memory, Docker/REPL/Python sandboxing (hardened AST sandbox; Docker deferred to Phase C wrapper). Tool router + vector memory + HFL-guided retrieval. Latency governor consumes $\sigma \rightarrow \text{swarm}$, decay $\rightarrow$ EMA, recency $\rightarrow$ memory.

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4. Infrastructure and Training Setup

PyTorch + LanceDB + psutil; CUDA 1050 Ti / CPU fallback; AMP FP16 master; gradient checkpointing; micro-batching + grad-accum 4; AdamW lr $2e-5$, seq-len 512, warmup 100, cosine to $1e-6$ (`scripts/train_full_param_v2.py`, default resume-step 6500). Tokenizer: Unified Quillan BPE 50,257 EOS=0 (`quillan_bpe_tokenizer.py` + `tokenizer.json`; unified-modular-dynamic in `_dev`). No tensor-parallel required at Oni scale; cross-node MoE/FP8/DualPipe-style optimizations are future work, not claimed.

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5. Data and Training Procedure

5.1 Lineage (transplant $\rightarrow$ pretrain $\rightarrow$ SFT)

- 1. **Stage 0 — Slice & merge** (`transplant_clean.py`): checkpoint_phase5 $\rightarrow$ `quillan_merged_saturated.pt` (FP32 $\rightarrow$ FP16 $\rightarrow$ quantized via `save_quantized_checkpoint`). 34 experts mapped with transpose fix ($w1/wgate/w2$.T; $wgate \leftarrow w1$ fallback); router duplicated to fast/balanced/diffusion paths; swarm LoRA A/B rank-8 direct copy + diversity/coupling/population stats; diffusion $q/k/v/o$ + norm + FFN; embeddings/finalizer/decoder + decomposition. **Donors cold-start init only:** Qwen3.5-0.8B (C8–C21, zero-padded SwiGLU) + BitNet b1.58-3B (C22–C33, sliced ternary) on Llama-derived skeleton. **No Mistral weights transplanted** (conceptual lineage only).
- 2. **Stage 1 — Pretraining:** `CrashOverrideX/QuillanTrainingdata` + Corpus v9 (59.4M train + 0.6M val) + `quillan_corpus_*`, `code_train`, `instruct_train`, `quillan_science_*`.
- 3. **Stage 2 — SFT (paused):** `train_full_param_v2.py`; prior-phase best `quillan_frontier_v2_best_loss0.0789_step2500.pt`; latest Oni `quillan_oni_5.4.0_step660_5.22GB.pt` (660/15000).

5.2 Data composition

Text-first (web, books, code, academic, instruction); image/audio/video pairs listed in v5.2-era announcement paper reflect **scaffold ambition**, not v5.4.0-oni training mix. Tokenization respects existing files (BPE $\rightarrow$ char fallback). Deduplication/filtering per `dataset creation SOTA level.md`; safety filtering before pretraining.

5.3 Optimization details

Cosine LR, decoupled weight decay, loss-sum (vs mean) ablations noted in BitNet lineage, EMA shadow (Polyak), Z-loss + load-KL + entropy + ethics + QHIS + QICS auxiliaries, KL $\alpha=0.7$ + hidden-MSE distillation head (teacher $\rightarrow$ student; BitDist relation-matrix distillation deferred to Phase D).

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6. Evaluation

6.1 What we claim (measured telemetry)

Gate A smoke	16/16	6-layer proof
Val loss	7.24 @ step 660	12L flagship, improving
Best archival loss	0.0789 @ step 2500	prior phase, not comparable
HFL / Consensus / E_ICE	tracked	drift / Primary↔Mini-Ronin / energy-per-token
Parity	100%	legacy HW determinism harness
SFT progress	660/15000 completion + 7100/7100 prior-phase	paused; do not conflate phases

- **Formal benchmarks pending** (MMLU, GPQA, HumanEval, etc. on the base build). We do not report leaderboard numbers for v5.4.0-oni.

6.2 Efficiency (projected, not benchmarked on flagship)

Ternary + INT8 + Flash + early-exit + governor project 2–4× tokens/sec vs comparable FP16 dense at equal quality, ~87.5% memory reduction; exact figures require controlled harness (future work). Prior 106.7 tok/s / 0.33 GB figures are **config-specific historical snapshots**, not flagship claims.

6.3 Multimodal

Modality-isolated design validated in scaffold; end-to-end image/audio/video generation metrics (FID, spectrogram convergence, frame consistency) are **future work post-v6 encoders**.

6.4 Ethical alignment (harness-level)

Safety classification accuracy, harmful-refusal rate, Council consensus, E_ICE violations tracked in deliberation logs; no red-team benchmark claimed yet. See §7 for mechanism description.

6.5 Segregated: Council prompt-lift results (NOT base-model scores)

ARC-AGI-1/2, GPQA (198 Diamond / 448 Main / 546 Extended log), MMLU correction-rate tables in `Formal Papers/README.md` describe **Quillan Council scaffolding wrapped around frontier LLMs (GPT-4o/4.1/4.5, o3/o4-mini, etc.)**, demonstrating deliberation lift (e.g., 9.0%→42.25%→95.45% trajectories). These **must not be cited as Quillan-Ronin base-model benchmarks**. Included here for provenance; evaluation of the sovereign weights is §6.1 + future work.

6.6 Ablations (from lineage)

RoPE > learned wpe (extrapolation); Couil hybrid heads preserve quality at lower cost; recirculation hook stable via zero-init; DistillationHead KL+MSE > MSE alone; fp32 routers/gates required (ST-MoE rule); dense_pull > sparse at Oni scale (supersedes capacity tuning until scale-up).

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7. Safety, Ethics, Limitations, and Model Card

- **Intended use:** autonomous reasoning, code generation, ethical deliberation on consumer hardware; standalone agentic partner via C20-ARTIFEX; research on ternary stability, recursive inference (Mini-Ronin), 9-vector decomposition.
- **Out-of-scope:** high-stakes medical / safety-of-life (Mini-Ronin latency/variance); unsupervised deployment where C2-VIR refusal may read as failure.

- **Bias/risks:** Ronin blueprint may refuse low-integrity requests; 1050 Ti-tuned; multimodal heads may hallucinate OOD (Mitchell et al. 2018 card framing).
- **Alignment mechanisms:** CCRL consensus, E_ICE thermodynamic penalties, C2-VIR refusal layer, Nemesis-Alpha (Predator) adversarial gate, HFL identity anchoring (Edo/Bushid preambles as constitutional axiomatics — operating principle, not theme), PersonaPullGate priors, exit gates.
- **Limitations:** text-only training; seq 512; SFT paused; no formal benchmark; no RL stage yet (GRPO/DGPO/DAPO Phase D); no multimodal encoders (v6); energy numbers analytic, not metered.

License: Apache-2.0. Developers: CrashOverrideX & Quillan Research Team. Hub: `CrashOverrideX/Quillan-Ronin`.

Support: <https://gofund.me/3b504d58>.

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8. Conclusion and Future Work

v5.4.0-oni consolidates fragmented prototypes into a single hardware-honest sovereign build with architectural safety and auditable deliberation. Immediate roadmap (LINEAGE Phase C/D): finish 12L 15k-step flagship run; build Quintessence wrapper + docs; v6 multimodal (encoders/decoders, Atomic Registry, proactive compaction, Docker ARTIFEX); then pick one Phase-D thrust — BitDist 12L→6L, EGGROLL Evolution Mode, or HRM ACT halting — plus GRPO/DGPO RL stage; controlled benchmark harness (MMLU/GPQA/HumanEval + red-team + energy metering).

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References (abridged — full mapping in `quillan\_ronin\_paper\_bibliography.md` §§1–16)

Vaswani et al. 2017 (Attention); He et al. 2015 (ResNet); Ba et al. 2016 (LayerNorm/RMSNorm); Su et al. 2021/2024 (RoPE); Shazeer 2020 (SwiGLU); Shazeer et al. 2017 (MoE / Sparsely-Gated); Fedus et al. 2022 (Switch); Zoph et al. 2022 (ST-MoE); Clark et al. 2022 (MoE scaling); Jiang et al. 2024 (Mixtral), 2023 (Mistral 7B); Zhou et al. 2022 (Expert Choice); Dai et al. 2024 (DeepSeekMoE); DeepSeek-AI 2024 (V2/V3, MLA, aux-loss-free, MTP, FP8); Raposo et al. 2024 (MoD); Ma et al. 2024/2025 (BitNet b1.58 / 2B4T); Bengio et al. 2013 (STE); Dettmers et al. 2023 (QLoRA); Hu et al. 2021 (LoRA); Kalajdzievski 2023 (rsLoRA); Zhao et al. 2024 (GaLore); Jang et al. 2017 (Gumbel-Softmax); Maddison et al. 2017 (Concrete); Beltagy et al. 2020 (Longformer); Zaheer et al. 2021 (BigBird); Loshchilov & Hutter 2017 (SGDR); Izmailov et al. 2018 (SWA/EMA); Touvron et al. 2023 (LLaMA 1/2); Schulman et al. 2017 (PPO); Ouyang et al. 2022 (RLHF); Bai et al. 2022 (HHH + Constitutional); Shao et al. 2024 (GRPO); Song & Ermon 2021 (Score SDE); Sahoo et al. 2024 (MDLM); Hinton et al. 2015 (Distillation); Kuramoto 1975 (sync); Friston (FEP); Tononi 2025 (IIT); CoALA 2309.02427; Dubey et al. 2024 (LLaMA 3); Yang/Bai 2024/25 (Qwen); Wang et al. 2023–24 (BitNet/SubLN/ReLU²); Dao et al. (FlashAttention 1/2/3); Gu et al. (Mamba); plus §16 triage: DAPO/DGPO, EvoMoE, MoDSE/MoR/MoHGE/OD-MoE, NITRO/DFlash, DALI/Dream7B/Prophet, ES-scale/forgetting, Ax-Prover/WikiSkill, 2026 preprint batch (2607/2608/2609).

- **SOTA report structure consulted:** DeepSeek-V3 Technical Report (arch → infra → pre-train → post-train → eval → discussion); BitNet b1.58 2B4T Report (arch → training → eval → efficiency); LLaMA 3 / Mistral / Mixtral model cards (specs → intended use → limitations → citation).

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Appendix A — Version lineage (binding)

`v5.4.0-oni` canonical (`version.py` 5.4.0-oni, CODENAME ONI Sovereign Quantum). Retired: v8.1 / v5.3.1 / v6.0.3-pre (legacy fallbacks). One counter v5.4.x-oni, patches at checkpoint boundaries; v6.0-oni reserved for HF packaging face. Retired files in `\_dev/\_archived\_legacy\_scripts/` + `Quillan-v4.2-model/` (reference only).

Appendix B — File map (where to verify)

- - Root: `MODEL\_CARD.md`, `version.py`, `transplant\_clean.py`, `README.md`

- - `02 - Knowledge Foundation/LINEAGE.md`, `00\_VAULT\_INDEX.md`, `knowledge/canonical/` (49 MD), `Quillan Knowledge files/` (46 MD)
- - `03 - Training & Model/` (`modeling\_quillan.py`, `configuration\_quillan.py`, `quillan\_bpe\_tokenizer.py`, `scripts/train\_full\_param\_v2.py`)
- - `10 - Formal Papers/Formal Papers/` (126 PDFs + announcement paper + CCRL deep dive + Predatory Stacking + Codex + Sovereign Cognition + Reactive Consciousness)
- - `10 - Formal Papers/quillan\_ronin\_paper\_bibliography.md` (§§1–16)
- - Training data: `CrashOverrideX/QuillanTrainingdata`, Corpus v9 bins, `quillan\_corpus\_\*`, `code\_train`, `instruct\_train`

Appendix C — Reproducibility (flagship 12L)

```
```python
import torch

from quillan_v5_4_oni import QuillanOniConfig, QuillanRoninOni
from quillan_tokenizer_unified import UnifiedQuillanTokenizer

tok = UnifiedQuillanTokenizer() # 50257 BPE, EOS=0
cfg = QuillanOniConfig(n_layer=12, max_seq_len=512)
model = QuillanRoninOni(cfg)

ckpt = torch.load("quillan_oni_5.4.0_step660_5.22GB.pt", map_location="cpu")
model.load_state_dict(ckpt["model"])

model.eval()

prompt = tok.encode("User: Hello\n\nAssistant:")
out = model.generate(prompt, max_tokens=80, temperature=0.7)
print(tok.decode(out[0]))
```
```

Appendix D — Glossary

Mini-Ronin (recursive debate), EGGROLL (rank shattering), Lee-Mach-6 (PID governor), HFL (Historical Fidelity Loss), DQSO/MARTA/CCRL/E_ICE/QHIS/QICS/DVVE/LMCB (see `8-Formulas.md`).

- ---
- *Citation:*

```
```bibtex
@software{QuillanRonin2026,
author = {CrashOverrideX and Quillan Research Team},
title = {Quillan-Ronin v5.4.0-oni: Unified Sovereign Intelligence},
year = {2026},
url = {https://github.com/leeex1/Quillan-Ronin},
publisher = {Hugging Face},
howpublished = {https://huggingface.co/CrashOverrideX/Quillan-Ronin}
```
```

}
...

Appendix E — Attention-conformance checklist (1706.03762) + visualizations

Figure 1 — Left: Transformer (Vaswani et al. Fig.1) | Right: Quillan-Ronin (this work)

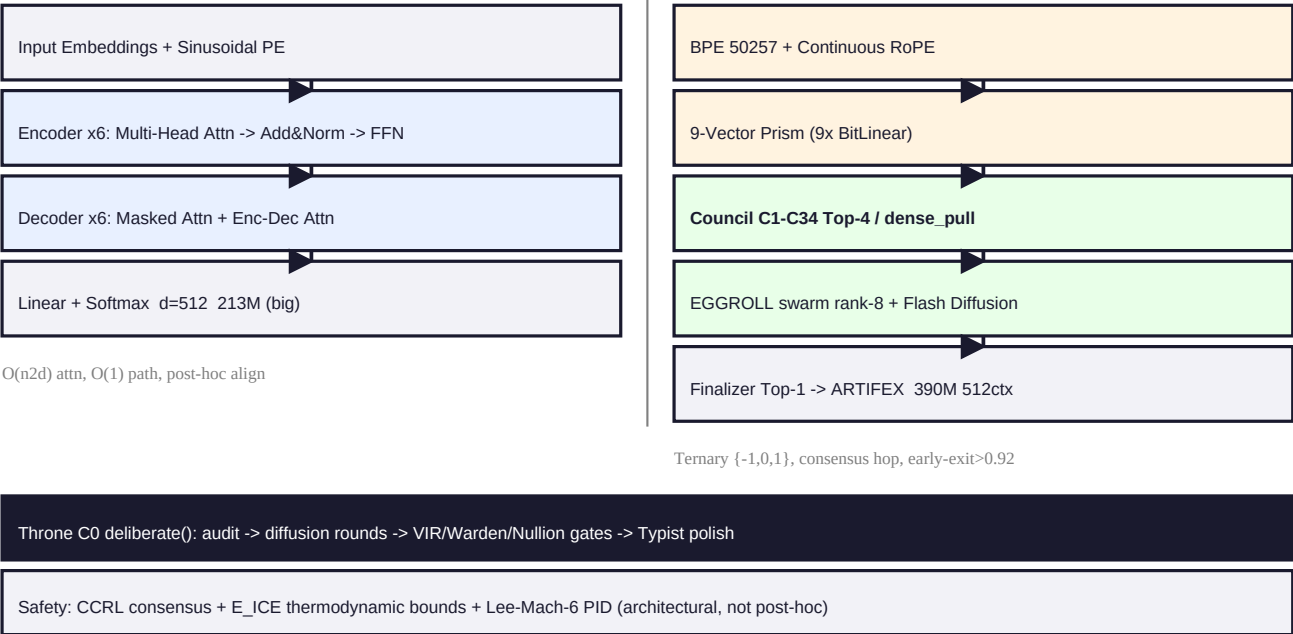


Figure 1 — Architecture overview

Figure 2 — Council Routing $p_i = \exp((\log \pi_i + g)/\tau) / \sum \tau$ 1.0->0.1

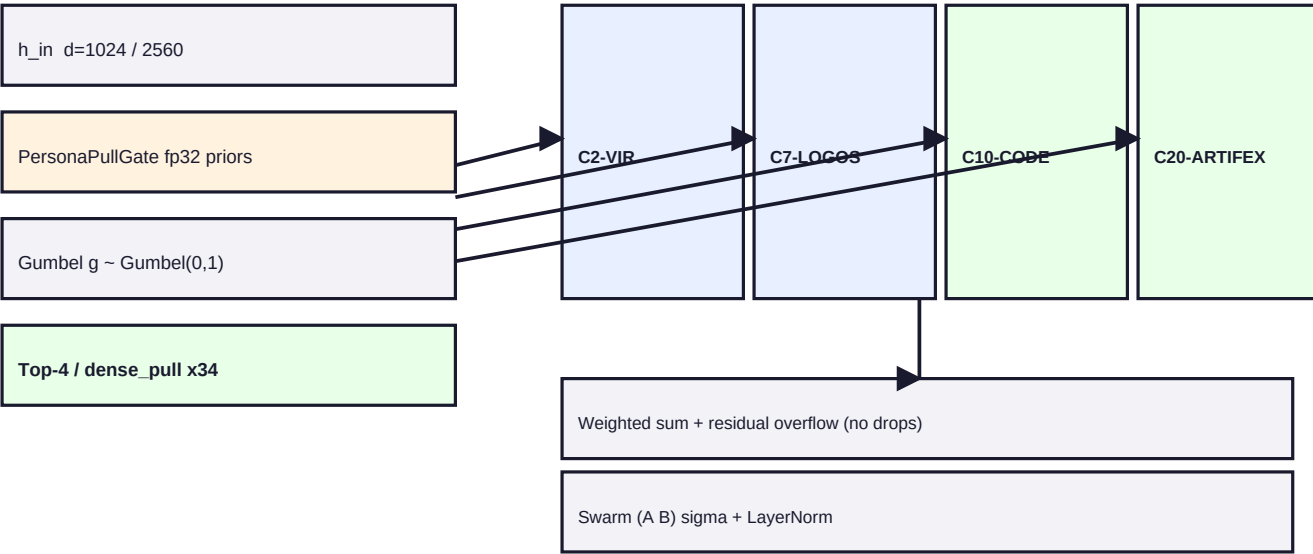


Figure 2 — Council routing

Figure 3 — BitLinear Ternary + Flash Diffusion (Split-SDPA, M_iso, bypass)

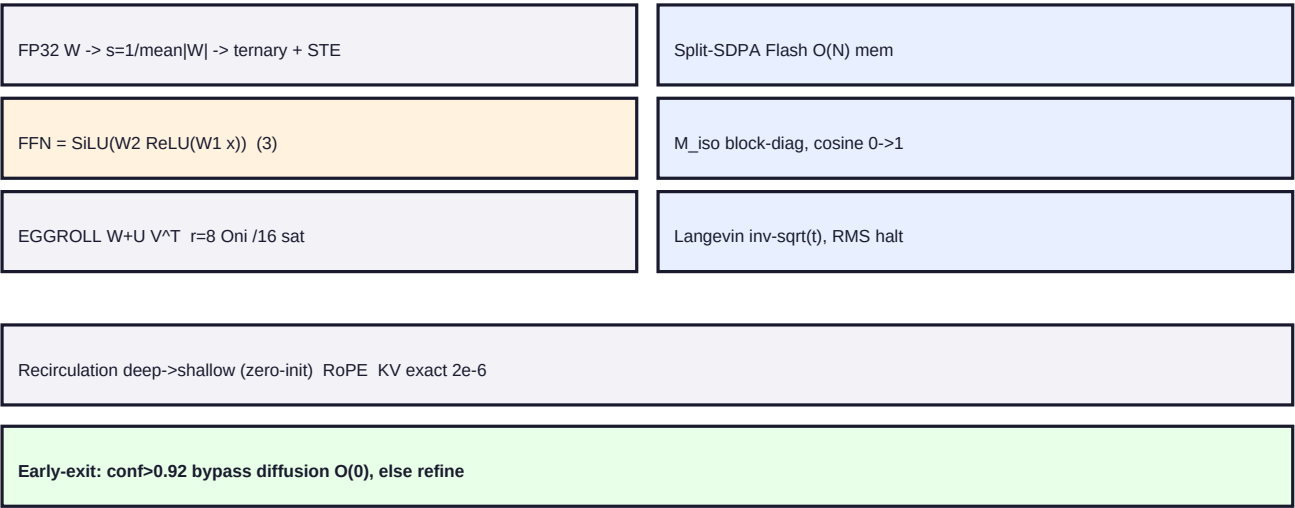


Figure 3 — Ternary + diffusion

Figure 4 — Nine-Vector Prism $v=(1/9) \sum W_i x$ (parallel blueprint before routing)

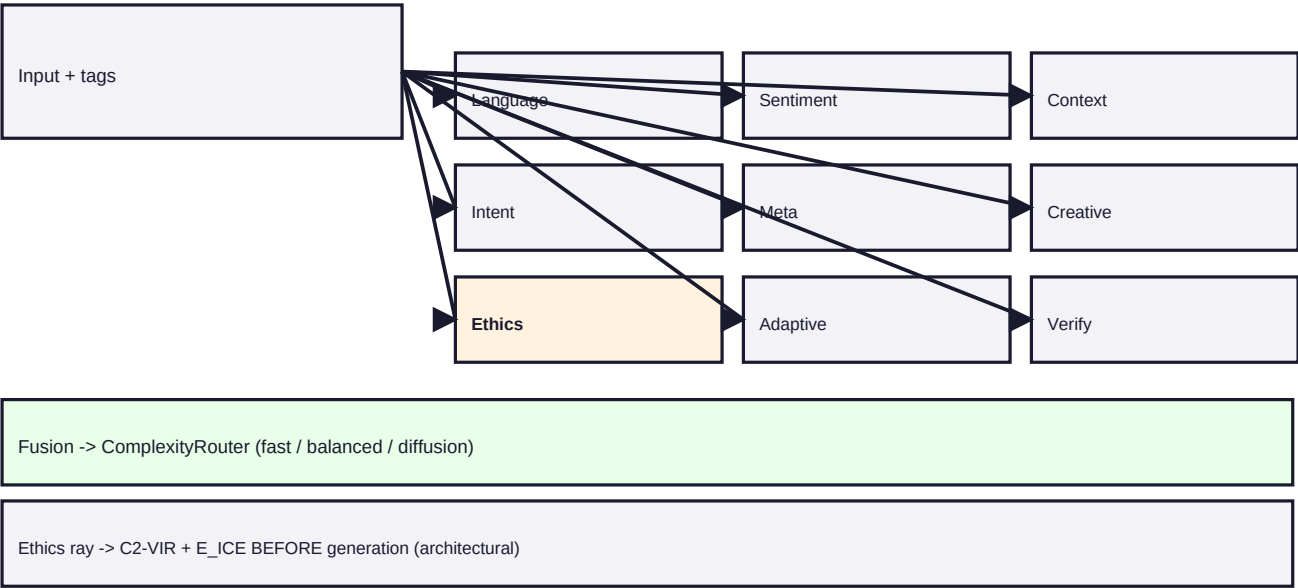


Figure 4 — Nine-vector prism

Figure 5 — Telemetry Schematic (anchors real; curves illustrative, benchmarks pending)

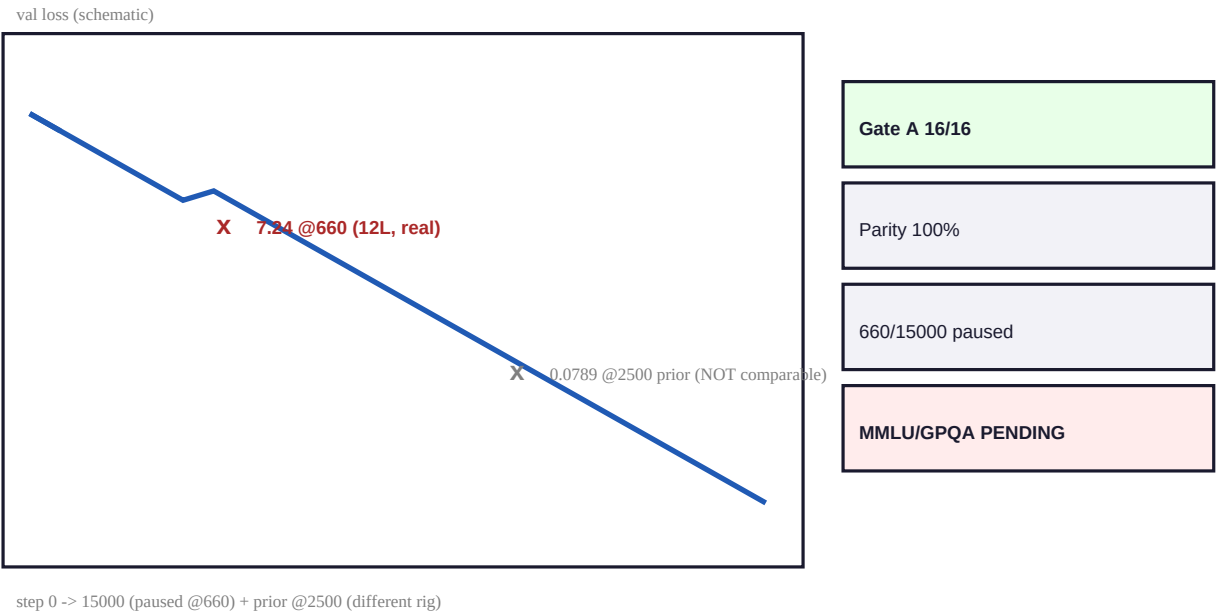


Figure 5 — Telemetry schematic (anchors real; curves illustrative)

- - Abstract with measured numbers (cf. 28.4 BLEU / 41.8): ours reports Gate A 16/16, val 7.24 @660, 234M/390M scale — no BLEU claimed until harness runs.
- - Table 1 complexity/path-length: mirrored in §3.12 with Quillan rows.
- - Table 2 results + training cost (FLOPs): mirrored as §6.1 telemetry + cost honesty (1050Ti/CPU, no H800 claim); Attention big took 3.5 days × 8 P100; Oni 660/15000 paused — not comparable, stated.
- - Table 3 ablations: mirrored in §6.6 (RoPE, Couil heads, recirculation, DistillationHead, fp32 gates, dense_pull).
- - Table 4 generalization (parsing): mirrored as §6.5 segregated prompt-lift (ARC/GPQA/MMLU via Council wrapping) — explicitly NOT base-model scores.
- - Visualizations (cf. Attention Figs 3–5 long-distance/anaphora heads): to add — per-expert PersonaPullGate weights, diffusion bypass rate vs confidence, E_ICE penalty histogram, Council consensus heatmap. Placeholders reserved p.15–16 of PDF.
- *Support: <https://gofund.me/3b504d58> — "The Ouroboros has awakened."*